Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** and JavaScript **API** services, multi-provider AI model routing with cost tracking, authentication and access control, and customer-facing features on AWS and Azure, with testing, code reviews, and monitoring.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **AI job orchestration** with FastAPI workers, explicit job-state workflows, and failure isolation, processing 1K+ LLM jobs at 3x throughput without cascading failures under concurrent load.
- Designed multi-tenant storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and **monitored** integration jobs to sustain 99%+ reliability through CI/CD releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **model routing** layer across OpenAI, Anthropic, and Qwen inference APIs with dynamic switching, timeouts, and fallbacks, serving 10K+ daily requests for 150+ instructors on AWS.
- Logged model name, feature, token counts, and estimated cost for every call, enabling **usage-based cost tracking** across grading, rubric, and OCR features while cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed **REST API** endpoints with pagination, field filtering, and caching for clients on low-bandwidth networks, improving response times 30% and cutting payload sizes across large-scale integrations used by distributed clients.
- Wrote 20+ automated unit and integration test suites with PyTest and Mockito, run nightly through Jenkins and GitHub-triggered branches across Windows, Linux, and macOS in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **authentication** with JWT, role-based access, and **billing** services in a JavaScript Node.js microservices platform, syncing 2M+ records across locations through real-time WebSocket updates for customer and lead workflows.
- Shipped through Jenkins CI/CD with Docker and staged releases, and debugged production issues with Sentry, Azure Monitor, and Kibana, cutting deployment time 50% and reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built customer-facing Flask APIs for bookings, pricing, and coupons with configurable eligibility rules, keeping responses under 200ms across 50K+ records with Firebase real-time updates on Android, iOS, and web.
- Integrated multi-currency payment providers through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and implemented wallet-signature login that removed password storage for wallet users.

Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, Bash
APIs & Backend: REST APIs, FastAPI, Flask, Node.js, Express.js, Microservices, Webhooks, JWT
AI Platform: LLM APIs, Model Routing, Cost Tracking, Prompt Engineering, Embeddings, PyTorch
Cloud & Ops: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Sentry, Kibana, Linux
Practices: Git, GitHub, Code Reviews, Unit Testing, PyTest, Debugging, Documentation, Agile, Postman

Projects

Inference Endpoint Control Plane | *Go, Python, REST APIs, Kubernetes, PostgreSQL*

- Designed a **control plane** API that lets developers create, scale, and delete model endpoints, stores desired configuration in Postgres, and continuously reconciles it with **Kubernetes** deployments and reports status back.

API Key and Usage Metering Service | *Python, FastAPI, Redis, PostgreSQL, OpenTelemetry*

- Designed a service that issues scoped **API keys**, rate-limits requests per key, records tokens and compute time for **billing**, and exposes per-project usage dashboards to customers and admins.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **customer-facing** platform on Kubernetes with CI/CD, Plaid bank linking, a token-based firewall, and real-time updates, supporting 50,000+ users and 1M+ transactions with MongoDB and Next.js.